Graphs of Sine and Cosine Functions

ANSWER KEY

Amplitude, period, and midline

1 For the function $y = 3\sin(2x)$, state:

a the amplitude $= 3$ **b** the period $= \pi$ **c** the range $[-3, 3]$

2 For the function $y = -2\cos\left(\frac{x}{2}\right) + 1$, state:

a the amplitude $= 2$ **b** the period $= 4\pi$

c the midline $y = 1$ **d** the range $[-1, 3]$

3 For $y = \sin\left(x - \frac{\pi}{3}\right)$, state the phase shift and the period.

Phase shift: $\frac{\pi}{3}$ to the right. Period: 2π.

Reading and writing equations from graphs

4 The graph below shows one sinusoidal function. State its amplitude and period, then write an equation for the function.

Amplitude 2, period 2π. Equation: $y = 2\sin x$.

5 Write an equation of a cosine function with amplitude 4, period π, reflected across the x-axis, with no phase or vertical shift.

$y = -4\cos(2x)$ (period $\frac{2\pi}{b} = \pi$ gives $b = 2$).

6 The depth of water at a harbour entrance is modeled by $d(t) = 2.5\sin\left(\frac{\pi}{6}t\right) + 7$ metres, where t is in hours after midnight.

a State the maximum and minimum depths. Maximum $7 + 2.5 = 9.5$ m; minimum $7 - 2.5 = 4.5$ m.

b State the period of the tide cycle. Period $= \frac{2\pi}{\pi/6} = 12$ hours.

c Find the depth at $t = 3$. $d(3) = 2.5\sin\left(\frac{\pi}{2}\right) + 7 = 9.5$ m.

Graphing with transformations

7 The graph below shows $y = \cos(x) + 1$, a cosine curve with a vertical shift. Identify the midline and the maximum and minimum values shown.

Midline $y = 1$; maximum value 2; minimum value 0.

8 Describe, in order, the transformations that turn $y = \sin x$ into $y = -3\sin\left(2\left(x - \frac{\pi}{4}\right)\right) + 2$.

Horizontal compression by a factor of 2 (period π), reflection across the x-axis, vertical stretch by a factor of 3, shift right $\frac{\pi}{4}$, and shift up 2.

9 State the amplitude, period, phase shift, and midline for $y = 5\cos\left(3x + \frac{\pi}{2}\right) - 4$. (Hint: factor the argument as $3\left(x + \frac{\pi}{6}\right)$.)

Amplitude 5; period $\frac{2\pi}{3}$; phase shift $\frac{\pi}{6}$ to the left; midline $y = -4$.

Modeling periodic phenomena

10 The average monthly temperature in a city varies between $8°$C (in January, month 1) and $28°$C (in July, month 7), following a sinusoidal pattern with a 12-month period.

a Find the amplitude and midline of the model. Amplitude $= \frac{28-8}{2} = 10$; midline $= \frac{28+8}{2} = 18$.

b Write a cosine model $T(m)$ for the temperature in month m, using the fact that the minimum occurs at $m = 1$. Since the minimum is at $m = 1$, use a negative cosine shifted:
$T(m) = -10\cos\left(\frac{2\pi}{12}(m-1)\right) + 18 = -10\cos\left(\frac{\pi}{6}(m-1)\right) + 18$.

c Use your model to estimate the temperature in month 4 (April).
$T(4) = -10\cos\left(\frac{\pi}{6}(3)\right) + 18 = -10\cos\left(\frac{\pi}{2}\right) + 18 = 18°$C.

Comparing sine and cosine graphs

11 Explain why $\cos x = \sin\left(x + \frac{\pi}{2}\right)$ by comparing the graphs of sine and cosine.

The cosine graph looks identical to the sine graph shifted $\frac{\pi}{2}$ units to the LEFT. Shifting $\sin x$ left by $\frac{\pi}{2}$ means replacing x with $x + \frac{\pi}{2}$, giving $\sin\left(x + \frac{\pi}{2}\right) = \cos x$.

12 Both $y = 2\sin(x)$ and $y = 2\cos(x)$ are graphed below. State one key visual difference and one similarity between them.

Similarity: both have the same amplitude (2) and period (2π). Difference: the sine curve passes through the origin $(0, 0)$, while the cosine curve reaches its maximum at $x = 0$ — they are horizontally shifted versions of the same shape.

Finding zeros and extrema from an equation

13 Find all zeros of $y = \sin(2x)$ on the interval $[0, 2\pi]$.

$2x = 0, \pi, 2\pi, 3\pi, 4\pi \Rightarrow x = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi$.

14 Find the x-value(s) in $[0, 2\pi]$ where $y = 3\cos(x)$ attains its maximum and minimum values.

Maximum value 3 occurs at $x = 0$ and $x = 2\pi$ (where $\cos x = 1$). Minimum value -3 occurs at $x = \pi$ (where $\cos x = -1$).

Solving basic sinusoidal equations

15 Solve $2\sin(x)+1=0$ for $x\in[0,2\pi]$.
$\sin x=-\frac{1}{2}$: $x=\frac{7\pi}{6}$ or $x=\frac{11\pi}{6}$.

16 A Ferris wheel's height above ground is $h(t)=15\sin\left(\frac{\pi}{10}t-\frac{\pi}{2}\right)+17$ meters. Find the first time $t>0$ (minutes) when the rider is at height 17 m and rising.
Height $=17$ (the midline) when $\sin\left(\frac{\pi}{10}t-\frac{\pi}{2}\right)=0$. The rider is rising through the midline when the argument equals 0: $\frac{\pi}{10}t-\frac{\pi}{2}=0\Rightarrow t=5$ minutes.